

Now  $\sqrt{x^2 + y^2}$  is the distance from  $(x, y)$  to  $(0, 0)$ . Thus, to make  $|f(x, y) - 0| < 0.001$ , for example, we need only require  $(x, y)$  be within 0.001 of  $(0, 0)$ . More generally, for any positive number  $u$ , no matter how small, we are sure that  $|f(x, y) - 0| < u$  whenever  $(x, y)$  is no farther than  $u$  from  $(0, 0)$ . This is what we mean by saying that the difference  $|f(x, y) - 0|$  can be made as small as we wish by choosing the distance to be sufficiently small. Thus, we conclude that

$$\lim_{(x,y) \rightarrow (0,0)} f(x, y) = \lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y}{x^2 + y^2} = 0.$$

Notice that since this limit equals  $f(0, 0)$ , the function  $f$  is continuous at  $(0, 0)$ .

- (b) Although the formula defining the function  $g$  looks similar to that of  $f$ , we saw in Example 1 that  $g$ 's behavior near the origin is quite different. If we consider points  $(x, 0)$  lying along the  $x$ -axis near  $(0, 0)$ , then the values  $g(x, 0)$  are equal to 1, while if we consider points  $(0, y)$  lying along the  $y$ -axis near  $(0, 0)$ , then the values  $g(0, y)$  are equal to 0. Thus, within any distance (no matter how small) from the origin, there are points where  $g = 0$  and points where  $g = 1$ . Therefore the limit  $\lim_{(x,y) \rightarrow (0,0)} g(x, y)$  does not exist, and thus  $g$  is not continuous at  $(0, 0)$ .

While the notions of limit and continuity look formally the same for one- and two-variable functions, they are somewhat more subtle in the multivariable case. The reason for this is that on the line (1-space), we can approach a point from just two directions (left or right) but in 2-space there are an infinite number of ways to approach a given point.

## Exercises and Problems for Section 12.6

### Exercises

In Exercises 1–6, is the function continuous at all points in the given region?

- $\frac{1}{x^2 + y^2}$  on the square  $-1 \leq x \leq 1, -1 \leq y \leq 1$
- $\frac{1}{x^2 + y^2}$  on the square  $1 \leq x \leq 2, 1 \leq y \leq 2$
- $\frac{y}{x^2 + 2}$  on the disk  $x^2 + y^2 \leq 1$
- $\frac{e^{\sin x}}{\cos y}$  on the rectangle  $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}, 0 \leq y \leq \frac{\pi}{4}$
- $\tan(xy)$  on the square  $-2 \leq x \leq 2, -2 \leq y \leq 2$
- $\sqrt{2x - y}$  on the disk  $x^2 + y^2 \leq 4$

In Exercises 7–11, find the limit as  $(x, y) \rightarrow (0, 0)$  of  $f(x, y)$ . Assume that polynomials, exponentials, logarithmic, and trigonometric functions are continuous.

- $f(x, y) = e^{-x-y}$
- $f(x, y) = x^2 + y^2$
- $f(x, y) = \frac{x}{x^2 + 1}$
- $f(x, y) = \frac{x + y}{(\sin y) + 2}$
- $f(x, y) = \frac{\sin(x^2 + y^2)}{x^2 + y^2}$  [Hint:  $\lim_{t \rightarrow 0} \frac{\sin t}{t} = 1$ .]

### Problems

In Problems 12–13, show that the function  $f(x, y)$  does not have a limit as  $(x, y) \rightarrow (0, 0)$ . [Hint: Use the line  $y = mx$ .]

- $f(x, y) = \frac{x + y}{x - y}, \quad x \neq y$
- $f(x, y) = \frac{x^2 - y^2}{x^2 + y^2}$
- By approaching the origin along the positive  $x$ -axis and the positive  $y$ -axis, show that the following limit does not exist:

$$\lim_{(x,y) \rightarrow (0,0)} \frac{2x - y^2}{2x + y^2}.$$

- Show that  $f(x, y)$  has no limit as  $(x, y) \rightarrow (0, 0)$  if

$$f(x, y) = \frac{xy}{|xy|}, \quad x \neq 0 \text{ and } y \neq 0.$$

- Show that the function  $f$  does not have a limit at  $(0, 0)$  by examining the limits of  $f$  as  $(x, y) \rightarrow (0, 0)$  along the curve  $y = kx^2$  for different values of  $k$ :

$$f(x, y) = \frac{x^2}{x^2 + y}, \quad x^2 + y \neq 0.$$

17. Let  $f(x, y) = \begin{cases} \frac{|x|}{x}y & \text{for } x \neq 0 \\ 0 & \text{for } x = 0. \end{cases}$

Is  $f(x, y)$  continuous

- (a) On the  $x$ -axis?      (b) On the  $y$ -axis?  
(c) At  $(0, 0)$ ?

In Problems 18–19, determine whether there is a value for  $c$  making the function continuous everywhere. If so, find it. If not, explain why not.

18.  $f(x, y) = \begin{cases} c + y, & x \leq 3, \\ 5 - x, & x > 3. \end{cases}$

19.  $f(x, y) = \begin{cases} c + y, & x \leq 3, \\ 5 - y, & x > 3. \end{cases}$

20. Is the following function continuous at  $(0, 0)$ ?

$$f(x, y) = \begin{cases} x^2 + y^2 & \text{if } (x, y) \neq (0, 0) \\ 2 & \text{if } (x, y) = (0, 0) \end{cases}$$

21. What value of  $c$  makes the following function continuous at  $(0, 0)$ ?

$$f(x, y) = \begin{cases} x^2 + y^2 + 1 & \text{if } (x, y) \neq (0, 0) \\ c & \text{if } (x, y) = (0, 0) \end{cases}$$

22. (a) Use a computer to draw the graph and the contour diagram of the following function:

$$f(x, y) = \begin{cases} \frac{xy(x^2 - y^2)}{x^2 + y^2}, & (x, y) \neq (0, 0), \\ 0, & (x, y) = (0, 0). \end{cases}$$

(b) Do your answers to part (a) suggest that  $f$  is continuous at  $(0, 0)$ ? Explain your answer.

23. The function  $f$ , whose graph and contour diagram are in Figures 12.89 and 12.90, is given by

$$f(x, y) = \begin{cases} \frac{xy}{x^2 + y^2}, & (x, y) \neq (0, 0), \\ 0, & (x, y) = (0, 0). \end{cases}$$

- (a) Show that  $f(0, y)$  and  $f(x, 0)$  are each continuous functions of one variable.  
(b) Show that rays emanating from the origin are contained in contours of  $f$ .  
(c) Is  $f$  continuous at  $(0, 0)$ ?

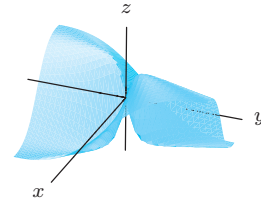


Figure 12.89: Graph of  $z = xy/(x^2 + y^2)$

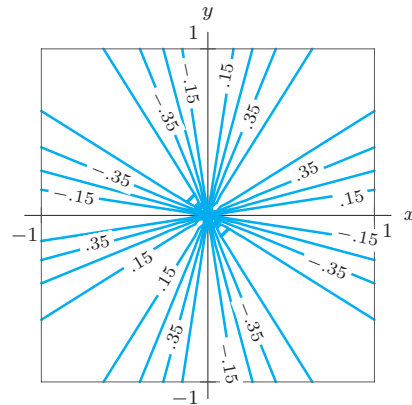


Figure 12.90: Contour diagram of  $z = xy/(x^2 + y^2)$

### Strengthen Your Understanding

In Problems 24–25, explain what is wrong with the statement.

24. If a function  $f(x, y)$  has a limit as  $(x, y)$  approaches  $(a, b)$ , then it is continuous at  $(a, b)$ .  
25. If both  $f$  and  $g$  are continuous at  $(a, b)$ , then so are  $f + g$ ,  $fg$  and  $f/g$ .

In Problems 26–27, give an example of:

26. A function  $f(x, y)$  which is continuous everywhere except at  $(0, 0)$  and  $(1, 2)$ .  
27. A function  $f(x, y)$  that approaches 1 as  $(x, y)$  approaches  $(0, 0)$  along the  $x$ -axis and approaches 2 as

$(x, y)$  approaches  $(0, 0)$  along the  $y$ -axis.

In Problems 28–30, construct a function  $f(x, y)$  with the given property.

28. Not continuous along the line  $x = 2$ ; continuous everywhere else.  
29. Not continuous at the point  $(2, 0)$ ; continuous everywhere else.  
30. Not continuous along the curve  $x^2 + y^2 = 1$ ; continuous everywhere else.